

JUNWON KIM

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RESEARCH INTERESTS

Machine Learning for Wireless Networks, Network Optimization and Resource Allocation, Multi-access Edge Computing, Reinforcement Learning and Transformer-based Models, 6G Non-Terrestrial Networks (NTN)

EDUCATION

YONSEI UNIVERSITY, Seoul, Republic of Korea *Mar. 2025 – Feb. 2027 (Expected)*
M.S. in Electrical and Electronic Engineering
Advisor: Prof. Jong-Moon Chung (Communications & Networking Lab.)

- Overall GPA: 4.11/4.3.

YONSEI UNIVERSITY, Seoul, Republic of Korea *Mar. 2019 – Feb. 2025*
B.S. in Electrical and Electronic Engineering

- Overall GPA: 3.7/4.3; Major GPA: 3.83/4.3.
- Including 1.5-year mandatory military service (Oct. 2020 – Apr. 2022).

PUBLICATIONS

Journal Articles

- [1] M. Choi, M. Park, **J. Kim**, and J.-M. Chung, “Intelligent Handover Scheme for Improved 6G NTN LEO Satellite Network Performance,” *IEEE Transactions on Mobile Computing*, vol. 25, no. 5, pp. 6863–6880, May 2026. DOI: 10.1109/TMC.2025.3642278.
- [2] **J. Kim**, M. Choi, and J.-M. Chung, “GT-PAR: Graph Transformer-Aided Traffic Prediction and Adaptive Routing for Dynamic LEO Satellite Networks,” *IEEE Wireless Communications Letters*, vol. 15, pp. 4290–4294, Jul. 2026. DOI: 10.1109/LWC.2026.3717034.

Conference Papers

- [1] **J. Kim** and J.-M. Chung, “Object Detection Robustness of UAV-MEC YOLO Split Computing under Signal Quality Degradation,” in *Proc. 2026 Spring Conf. of the Korea Institute of Electronic Communication Sciences (KIECS)*, Republic of Korea, Jun. 2026. **(Outstanding Paper Award)**

HONORS AND SCHOLARSHIPS

Outstanding Presentation Award, 8th Industry–Academia Project Showcase, Future Mobility Program, Yonsei University *Jul. 2026*

Outstanding Paper Award, 2026 Spring Conference of the Korea Institute of Electronic Communication Sciences (KIECS) *Jun. 2026*

Teaching Assistant Scholarship, Yonsei University *Fall 2025, Spring 2026*

RESEARCH EXPERIENCE

Communications & Networking Laboratory (CNL), Yonsei University *Mar. 2025 – Present*

Graduate Research Assistant (Advisor: Prof. Jong-Moon Chung)

Learning-based handover and routing for 6G NTN/LEO satellite networks

- Designed reinforcement-learning handover policies for LEO satellite networks, evaluated through large-scale network simulation.
- Developed GT-PAR, a graph-transformer framework for joint inter-satellite traffic prediction and adaptive routing in dynamic LEO networks (first author).

Communications & Networking Laboratory (CNL), Yonsei University

Jul. 2024 – Feb. 2025

Research Intern (Advisor: Prof. Jong-Moon Chung)

Wireless communications, UAV/satellite networking, and edge/cloud computing

- Investigated learning-based network optimization for satellite and UAV networks, implementing and benchmarking handover and routing schemes through network simulation.

Robotic & Mobile Networks Laboratory (RAMO), Yonsei University

Jan. 2024 – Jun. 2024

Research Intern, Undergraduate Thesis (Advisor: Prof. Seonglyun Kim)

Collaborative computing for URLLC in remote robot control over a private 5G system

- Built a multi-access edge computing (MEC) testbed offloading robot vision inference to an edge/core server over a private 5G network for ultra-reliable low-latency communication (URLLC) in remote robot control.
- Integrated vision-based object detection and monocular depth estimation on the edge server, and implemented a real-time safety mechanism that halts and resumes the robot upon object intrusion.

SELECTED PROJECTS

Adaptive UAV-MEC Collaborative Inference for Urban Traffic Monitoring,

Mar. 2026 – Jul. 2026

Yonsei University

Connected Mobility Research Project (Graduate Course)

- Built a UAV-MEC split-computing pipeline for real-time object detection in traffic monitoring, robust under poor wireless channels.
- Received the Outstanding Presentation Award at the 8th Industry-Academia Project Showcase, Future Mobility Program, Yonsei University (Jul. 2026).

UAV-Based Dynamic Object Recognition for Digital Land-Information Mapping,

Jan. 2025 – Present

Yonsei University

Research Assistant, Ministry of Land, Infrastructure and Transport (MOLIT)-funded national R&D program

- Built a deep learning-based detection system for real-time recognition of road objects from UAV aerial imagery.
- Improved detection of small, occluded, and low-light objects via tiling-based augmentation and misalignment-tolerant EO/IR sensor fusion, meeting accuracy and latency targets in official certification testing.

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Electrical and Computer Engineering (ECE 101)

Mar. 2026 – Jun. 2026

Yonsei University, School of Electrical and Electronic Engineering

- Assisted a 370-student introductory course on emerging AI, networking, and security technologies; handled grading, office hours, exam logistics, and student communication.

Teaching Assistant, Experiments on Communication Networks*Sep. 2025 – Dec. 2025*

Yonsei University, School of Electrical and Electronic Engineering

- Led hands-on labs for 20 undergraduates on packet analysis (Wireshark), spectrum sensing with CNN detection (USRP/LabVIEW), SDN routing (Mininet), and LTE/V2X simulation (OMNeT++); graded lab reports with individual feedback.

LEADERSHIP AND SERVICE

Mandatory Military Service: Training Assistant & Squad Leader*Oct. 2020 – Apr. 2022*

Republic of Korea Army Signal School

- Supervised squad-level activities and assisted communications training in a structured military environment.

President, POS Soccer Club, School of Electrical and Electronic Engineering,
Yonsei University*Jan. 2020 – Sep. 2020*

- Led club operations, member coordination, and activity planning.

TECHNICAL SKILLS AND CERTIFICATIONS

Programming

Python, C/C++, MATLAB, Verilog.

ML & Simulation

PyTorch; OMNeT++, SimuLTE, Veins/SUMO, Mininet, Wireshark.

Certification

Class III Unmanned Multirotor Pilot Certificate, Republic of Korea.

Languages

Korean (native); English.